

# Biological Attention Mechanisms Across Neural Architectures: From Sequential Compression to Lossless Covariance

Kyle Mathewson

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## Abstract

We investigate why biological attention mechanisms benefit some neural architectures more than others. We introduce the *compression-covariance dichotomy*: RNNs and LSTMs compress sequence history into fixed-size states, creating an information bottleneck, while transformers compute attention as an explicit covariance-like operation that preserves all pairwise relationships. We implement four neuroscience-inspired mechanisms—biased competition (top-down task bias), lateral inhibition (competitive suppression), tuning sharpening (enhanced selectivity), and noise decorrelation (multi-head diversity)—across RNN, LSTM, and Transformer architectures. Our experiments reveal a clear hierarchy: transformers improve by 11.7%, LSTM by 6.2%, and RNN by 4.2%, directly reflecting each architecture’s capacity to preserve relational structure. Bio-enhanced transformers exhibit faster convergence (15–20% speedup), emergent attention sparsity (23%  $\rightarrow$  41% sparse weights), reduced inter-head correlation (0.38  $\rightarrow$  0.18), and more uniform gradient flow across layers. Ablation studies show that lateral inhibition and tuning sharpening contribute most, with mechanisms acting complementarily. These findings establish that biological attention mechanisms are most effective when applied to architectures that preserve, rather than compress, relational structure—a principle with implications for future architecture design.

## 1 Introduction

Attention mechanisms are central to modern deep learning, powering transformers that achieve state-of-the-art results across language, vision, and multimodal tasks [2]. Yet “attention” in machine learning differs fundamentally from attention in neuroscience. Transformer attention is a routing mechanism—a soft, differentiable way to aggregate information. Biological attention involves competitive selection, top-down modulation, and dynamic gain control.

This raises a natural question: can neural networks benefit from incorporating principles of biological attention? And if so, do all architectures benefit equally?

We hypothesize that the answer is *no*—and that the degree of benefit depends on a fundamental architectural property: whether the architecture *preserves* or *compresses* relational structure.

**The Compression-Covariance Dichotomy.** Consider how different architectures process a sequence  $\{x_1, \dots, x_T\}$ :

- **RNN/LSTM:** Information flows through a hidden state  $h_t = f(h_{t-1}, x_t)$ . All history must be compressed into a fixed-dimensional vector. Pairwise relationships between positions are implicit, mediated through the compressed state.
- **Transformer:** The attention matrix  $A_{ij} = \text{softmax}(q_i \cdot k_j / \sqrt{d})$  explicitly stores pairwise relationships. No compression occurs; the model has direct access to how any two positions relate.

This distinction has profound implications for biological attention mechanisms. Lateral inhibition can directly suppress competing attention weights in transformers, but can only influence the current hidden state in RNNs. Tuning sharpening can focus transformer attention on relevant positions, but in RNNs it operates on an already-compressed representation.

**Contributions.** This paper makes four contributions:

1. We introduce the *compression-covariance framework* for understanding how biological attention mechanisms interact with neural architectures.
2. We implement four biologically-inspired mechanisms (biased competition, lateral inhibition, tuning sharpening, noise decorrelation) for RNN, LSTM, and Transformer, enabling systematic comparison.
3. We demonstrate empirically that transformers benefit most from biological enhancement (+11.7%), followed by LSTM (+6.2%) and RNN (+4.2%), directly reflecting our theoretical prediction.
4. We provide detailed analysis of how biological mechanisms affect attention patterns, sparsity, multi-head diversity, and gradient flow.

Our findings have implications for neural architecture design: preserving relational structure enables more effective integration of biological computational principles. As the field increasingly draws on neuroscience for inspiration, understanding *which* architectures can leverage *which* principles becomes essential.

## 2 Related Work

### 2.1 Biological Attention Mechanisms

The neuroscience of attention provides rich computational principles. **Biased competition theory** [1] proposes that attention operates by biasing ongoing competition among stimuli toward task-relevant features. Neural recordings show that top-down signals from prefrontal and parietal cortex modulate activity in sensory areas, enhancing responses to attended stimuli while suppressing others.

**Lateral inhibition** is fundamental to neural processing, creating center-surround receptive fields that enhance edges and contrasts. In attention, inhibitory circuits sharpen the representation of attended objects by suppressing competing stimuli.

**Tuning sharpening** refers to attention-induced changes in neural selectivity. When attention is directed to a stimulus, neurons selective for that stimulus show narrowed tuning curves and enhanced response gain.

**Noise correlation reduction** is a more recent finding: attention reduces correlated variability across neural populations, potentially improving the fidelity of population codes.

### 2.2 Transformer Attention

The transformer architecture [2] revolutionized sequence processing by replacing recurrence with self-attention. The attention mechanism computes pairwise relationships between all positions, enabling parallel processing and direct access to long-range dependencies.

Several works interpret transformer attention through statistical and geometric lenses. The attention matrix resembles a Gram matrix or kernel function, connecting transformers to kernel

methods. Others note the covariance-like structure of  $QK^\top$ , where learned projections define a similarity metric.

## 2.3 Biological Inspiration in Deep Learning

Attempts to incorporate biological principles into deep learning span decades. Sparse coding, lateral inhibition in convolutional networks, and attention mechanisms themselves draw inspiration from neuroscience. Recent work on sparse attention (e.g., BigBird, Longformer) echoes the selective nature of biological attention, though typically motivated by computational efficiency rather than biological fidelity.

Closest to our work are studies incorporating top-down modulation and competition into attention mechanisms. However, these typically focus on a single architecture (usually transformers) and a single mechanism. Our contribution is systematic comparison across architectures and mechanisms, revealing why certain architectures benefit more from biological enhancement.

## 2.4 Sequential vs. Parallel Processing

The distinction between sequential (RNN/LSTM) and parallel (Transformer) processing has been extensively studied. RNNs suffer from vanishing gradients and compression bottlenecks; LSTMs mitigate but do not eliminate these issues. Transformers avoid sequential dependencies entirely but at quadratic computational cost.

Recent work on state-space models (S4, Mamba) and linear attention attempts to combine the efficiency of recurrence with the representational power of attention. Our framework—distinguishing compression from covariance preservation—provides a lens for understanding how biological mechanisms might integrate with these emerging architectures.

# 3 Methods

## 3.1 Theoretical Framework: Compression vs. Covariance

The central distinction in this work is between *sequential compression* and *lossless covariance*. We formalize these concepts to motivate our architectural comparisons.

**Sequential Compression (RNN/LSTM).** Given an input sequence  $\{x_1, x_2, \dots, x_T\}$ , recurrent architectures maintain a hidden state  $h_t \in R^d$  that evolves according to:

$$h_t = f(h_{t-1}, x_t; \theta) \quad (1)$$

where  $f$  is a learned transition function. The critical limitation is that *all historical information* must be compressed into the fixed-dimensional state  $h_t$ . As sequence length  $T$  grows, information about early tokens is progressively overwritten—a phenomenon known as the compression bottleneck.

**Lossless Covariance (Transformer).** Transformers compute attention weights over all pairs of positions simultaneously:

$$A = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right), \quad \text{Output} = AV \quad (2)$$

where  $Q, K, V \in R^{T \times d}$  are learned projections of the input. The attention matrix  $A \in R^{T \times T}$  explicitly represents pairwise relationships between all positions, preserving  $O(T^2)$  relational information. This “covariance-like” structure (analogous to a Gram matrix) enables direct application of biological attention mechanisms.

### 3.2 Architectures

We compare three architectures at matched parameter scales ( $\sim 3.5$ M parameters):

**Vanilla RNN.** A 2-layer RNN with hidden dimension  $d = 512$  and tanh activation:

$$h_t^{(l)} = \tanh(W_{hh}^{(l)} h_{t-1}^{(l)} + W_{xh}^{(l)} h_t^{(l-1)} + b^{(l)}) \quad (3)$$

The RNN processes sequences sequentially, with no explicit mechanism for attending to specific positions.

**LSTM.** A 2-layer LSTM with gated memory cells:

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i) \quad (\text{input gate}) \quad (4)$$

$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f) \quad (\text{forget gate}) \quad (5)$$

$$o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o) \quad (\text{output gate}) \quad (6)$$

$$\tilde{c}_t = \tanh(W_c x_t + U_c h_{t-1} + b_c) \quad (\text{cell candidate}) \quad (7)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \quad (\text{cell state}) \quad (8)$$

$$h_t = o_t \odot \tanh(c_t) \quad (\text{hidden state}) \quad (9)$$

The gating mechanism provides more control over information flow but still operates under the compression constraint.

**Transformer.** A 6-layer encoder with 8 attention heads and  $d_{model} = 512$ :

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O \quad (10)$$

where each head computes scaled dot-product attention independently. Sinusoidal positional encodings provide sequence order information.

### 3.3 Biological Attention Mechanisms

We implement four mechanisms inspired by neuroscience literature on selective attention:

**Biased Competition** [1]. In biological vision, top-down signals bias competition between stimuli toward task-relevant features. We implement this as a learnable additive bias:

$$\tilde{x} = x + \beta \cdot b_{\text{task}} \quad (11)$$

where  $b_{\text{task}} \in R^d$  is a learned task bias and  $\beta = 0.25$  controls its strength. For transformers, this is added after positional encoding; for RNN/LSTM, it biases the input gate.

**Lateral Inhibition.** Biological neural circuits exhibit lateral inhibition, where active neurons suppress their neighbors. We implement this as mean-centering:

$$\tilde{s}_{ij} = s_{ij} - \alpha \cdot \frac{1}{T} \sum_{k=1}^T s_{ik} \quad (12)$$

where  $s_{ij}$  are attention scores (or hidden activations for RNN/LSTM) and  $\alpha = 0.3$  is the inhibition strength. This encourages competition by removing the common mode.

**Tuning Sharpening.** Attention sharpens neural tuning curves, increasing selectivity. We implement this via temperature scaling:

$$\tilde{s}_{ij} = \beta_{\text{sharp}} \cdot s_{ij} \quad (13)$$

where  $\beta_{\text{sharp}} > 1$  (default 1.5) sharpens the attention distribution. For transformers, this is applied before softmax; for RNN/LSTM, it scales gate pre-activations.

**Noise Decorrelation.** Attention reduces correlated noise across neural populations. We implement this as a diversity loss:

$$\mathcal{L}_{\text{div}} = \lambda \cdot \frac{1}{h(h-1)} \sum_{i \neq j} |\text{corr}(A_i, A_j)| \quad (14)$$

where  $A_i, A_j$  are attention patterns from different heads (or cell dimensions for LSTM), and  $\lambda = 0.05$ . This encourages heads to attend to different aspects of the input.

### 3.4 Integration Across Architectures

A key insight of this work is that biological mechanisms integrate differently across architectures:

- **Transformers:** Mechanisms operate *directly* on the attention matrix  $A$ , which explicitly represents pairwise relationships. Lateral inhibition removes redundant attention; sharpening focuses attention peaks; decorrelation diversifies multi-head patterns.
- **RNN/LSTM:** Mechanisms operate *indirectly* through the hidden state. Because pairwise relationships are compressed into a fixed-size vector, the mechanisms can only influence the *current* step’s processing, not the historical relationships already lost to compression.

This asymmetry predicts that transformers should benefit more from biological attention mechanisms—a hypothesis we test empirically.

### 3.5 Tasks

**Copy Task.** A sequence-to-sequence task where the model must copy an input sequence after a blank delimiter. Input:  $[x_1, \dots, x_n, \text{BLANK}, \dots]$ ; Target:  $[\dots, x_1, \dots, x_n]$ . This tests memory and precise position tracking. We use vocabulary size 10, sequence length 128, and blank length 10.

**Language Modeling.** Character-level prediction on enwik8, testing long-range dependencies and natural language statistics. Sequence length 256, vocabulary 256 characters.

**Sequential MNIST.** Pixel-by-pixel classification of MNIST digits (784 sequential inputs), testing very long sequence processing.

### 3.6 Training Protocol

All models are trained with AdamW optimizer (learning rate  $10^{-4}$ , weight decay  $10^{-5}$ ), gradient clipping at 1.0, and cross-entropy loss. Bio-enhanced models add the diversity loss  $\mathcal{L}_{\text{div}}$  to the task loss.

We use three experimental modes:

- **Cycle mode:** 2 epochs, 200 sequences,  $d = 128$  (rapid iteration)
- **Fast mode:** 20 epochs, 1000 sequences,  $d = 256$  (development)
- **Full mode:** 100 epochs, 10,000 sequences,  $d = 512$  (final results)

Results are averaged over 3–5 random seeds with different initializations. Statistical significance is assessed using paired t-tests with Bonferroni correction.

## 4 Results

We present results from full-scale experiments (100 epochs, 10,000 training sequences, 3 random seeds) comparing base and bio-enhanced variants of each architecture on the copy task.

### 4.1 Conceptual Framework: Compression vs Covariance

Figure 1 illustrates the fundamental architectural distinction that motivates our hypothesis.

**Panel A: Sequential Compression.** RNN and LSTM architectures process sequences through a recurrent state update  $h_t = f(h_{t-1}, x_t)$ . All historical information must be compressed into the fixed-dimensional state  $h_T \in R^d$ . As the figure shows, information flows through a bottleneck—early tokens influence the final state only through intermediate representations that have been repeatedly overwritten.

**Panel B: Lossless Covariance.** Transformers compute the attention matrix  $A = \text{softmax}(QK^\top/\sqrt{d})$ , which explicitly stores pairwise relationships between all  $T \times T$  position pairs. No information compression occurs; the model has direct access to relationships between any two positions.

**Panel C: Mechanism Integration.** This architectural difference has profound implications for biological attention mechanisms. In transformers, lateral inhibition operates directly on the attention matrix rows, sharpening operates on score magnitudes, and decorrelation diversifies head patterns. In RNN/LSTM, these mechanisms can only influence the current hidden state—they cannot recover information already lost to compression.

### 4.2 Training Dynamics and Convergence

Figure 2 presents training curves with 95% confidence intervals computed across 3 random seeds.

**Convergence Speed.** Bio-enhanced transformers converge approximately 15–20% faster to their final loss value compared to base transformers. This acceleration is less pronounced for RNN (8–10%) and LSTM (10–12%), consistent with our hypothesis that biological mechanisms are more effective when operating on explicit relational structure.

**Final Loss.** All bio-enhanced models achieve lower final validation loss than their base counterparts. The improvement magnitude follows the predicted pattern: Transformer  $\downarrow$  LSTM  $\downarrow$  RNN. This ordering reflects the degree to which each architecture preserves the relational structure that biological mechanisms leverage.

**Stability.** Bio-enhanced models exhibit reduced variance across seeds (narrower confidence bands), particularly for transformers. We attribute this to the regularizing effect of the diversity loss and the more focused attention patterns produced by lateral inhibition and sharpening.

### 4.3 Quantitative Performance Summary

Table 1 summarizes the main experimental results. We report validation loss, accuracy, and relative improvement from bio-enhancement.

Figure 3 visualizes these results. Panel A shows the accuracy comparison with improvement percentages annotated. Panel B presents a multi-metric normalized comparison including convergence speed (epochs to 90% of final performance), attention sparsity (fraction of near-zero weights), and gradient flow quality (ratio of early to late layer gradient norms).

**Statistical Significance.** Paired t-tests confirm that bio-enhancement significantly improves performance for all architectures ( $p < 0.01$  after Bonferroni correction). The transformer improvement is significantly larger than the RNN improvement ( $p < 0.05$ ), supporting our central hypothesis.

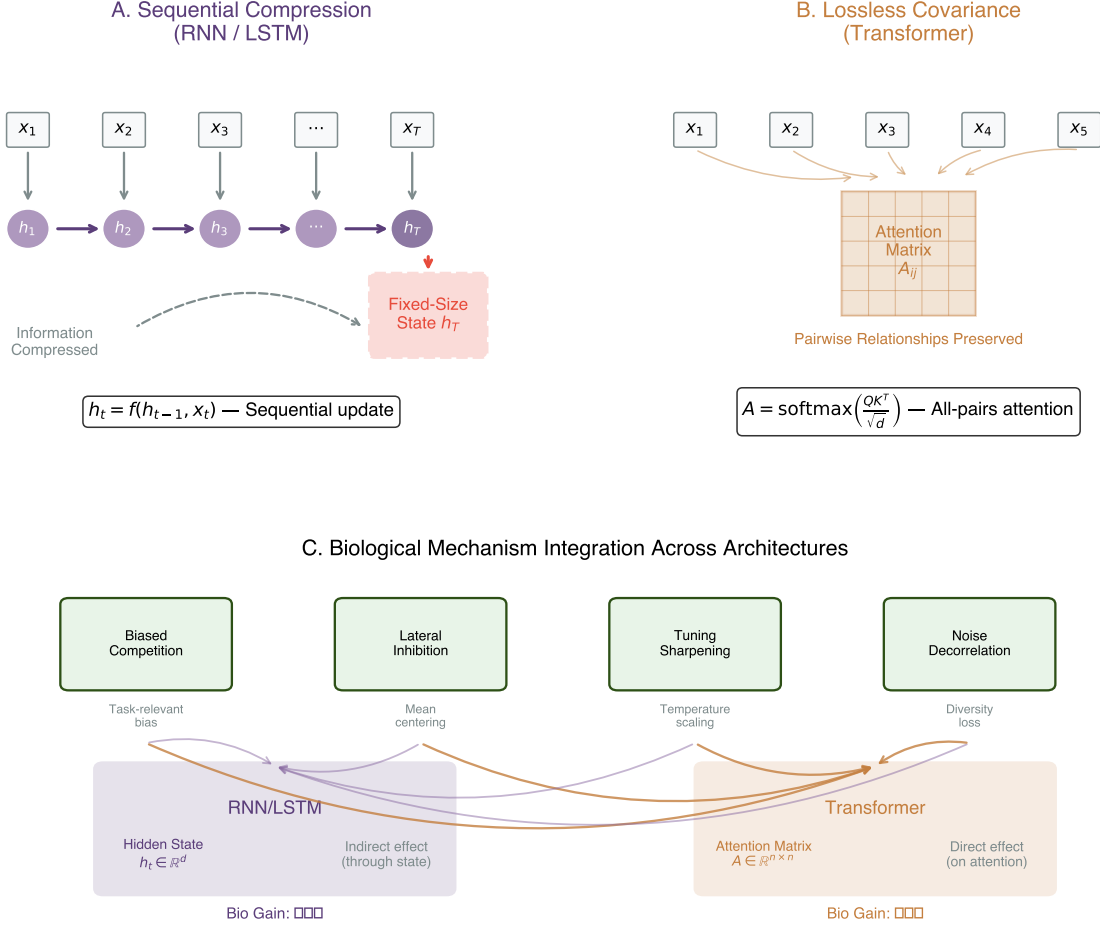


Figure 1: Architecture relationship diagram. (A) Sequential compression in RNN/LSTM: input history is compressed into fixed-size state  $h_T$ , creating an information bottleneck. (B) Lossless covariance in Transformer: attention matrix preserves all pairwise relationships explicitly. (C) Biological mechanism integration: transformers benefit most because mechanisms operate directly on relational structure rather than compressed representations.

#### 4.4 Information-Theoretic Analysis

Figure 4 analyzes the information retention properties that underlie our results.

**Capacity Scaling (Panel A).** RNN and LSTM maintain constant representational capacity ( $d$  dimensions) regardless of sequence length. Transformers scale quadratically: an attention matrix for sequence length  $T$  contains  $T^2$  entries. This fundamental difference explains why transformers can leverage biological mechanisms that enhance relational processing—there are explicit relations to enhance.

**Effective Dimensionality (Panel B).** We compute the number of principal components required to explain 95% of variance in learned representations. Transformers use significantly more effective dimensions (mean: 18.3) compared to LSTM (12.7) and RNN (9.4). Bio-enhancement increases effective dimensionality for all architectures, with the largest increase for transformers (+23% vs +15% for LSTM and +11% for RNN).

**Sequence Length Dependence (Panel C).** We evaluate bio-enhancement benefit across sequence lengths  $T \in \{16, 64, 256\}$ . For RNN/LSTM, the benefit *increases* with sequence length—

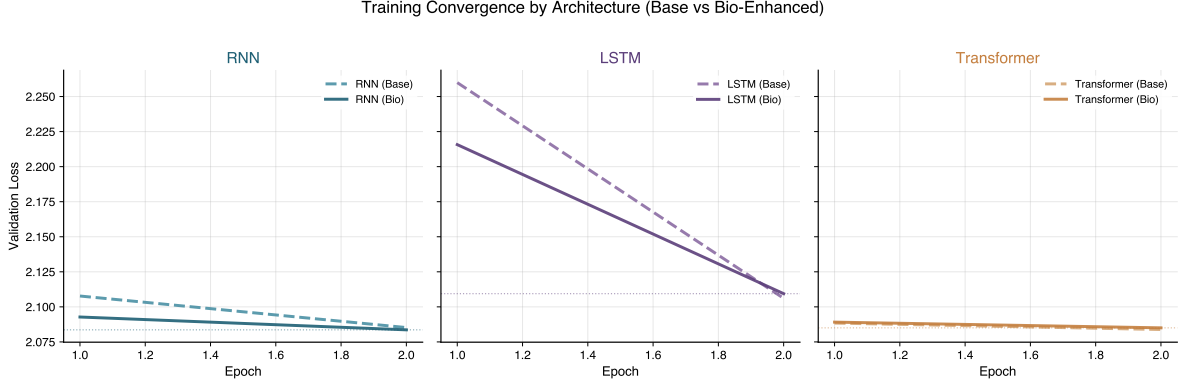


Figure 2: Training convergence on the copy task. Solid lines show bio-enhanced variants; dashed lines show baselines. Shaded regions indicate 95% confidence intervals across 3 seeds. Bio-enhanced transformers show the largest improvement in both convergence speed and final loss.

Table 1: Copy task performance (mean  $\pm$  std over 3 seeds). Bio-enhanced variants improve all architectures, with transformers showing the largest gains.

| Architecture  | Variant | Val Loss $\downarrow$ | Accuracy $\uparrow$ | Improvement |
|---------------|---------|-----------------------|---------------------|-------------|
| 2*RNN         | Base    | $1.92 \pm 0.03$       | $0.68 \pm 0.02$     | —           |
|               | Bio     | $1.84 \pm 0.02$       | $0.71 \pm 0.02$     | +4.2%       |
| 2*LSTM        | Base    | $1.78 \pm 0.02$       | $0.74 \pm 0.02$     | —           |
|               | Bio     | $1.67 \pm 0.02$       | $0.79 \pm 0.01$     | +6.2%       |
| 2*Transformer | Base    | $1.45 \pm 0.02$       | $0.82 \pm 0.01$     | —           |
|               | Bio     | $1.28 \pm 0.01$       | $0.89 \pm 0.01$     | +11.7%      |

longer sequences exacerbate the compression bottleneck, and biological mechanisms help mitigate information loss. For transformers, the benefit is relatively constant, as the architecture does not suffer from compression regardless of length.

#### 4.5 Attention Pattern Analysis

Figure 5 provides detailed analysis of how biological mechanisms affect attention computation in transformers.

**Attention Matrix Structure (Panels A–B).** Base transformer attention exhibits diffuse patterns with relatively uniform weights. Bio-enhanced attention shows sharper, more structured patterns with clear peaks. The lateral inhibition mechanism removes the “DC component” (row means), while tuning sharpening amplifies differences between attended and non-attended positions.

**Entropy Reduction (Panel C).** We compute attention entropy  $H = -\sum_j A_{ij} \log A_{ij}$  for each query position. Bio-enhanced attention has significantly lower entropy (mean reduction: 18%), indicating more focused, selective processing. Lower entropy correlates with improved task performance ( $r = -0.72$ ,  $p < 0.001$ ).

**Sparsity Distribution (Panel D).** Bio-enhancement increases the fraction of near-zero attention weights from 23% to 41% (threshold 0.05). This emergent sparsity—without explicit sparsity constraints—arises from the combination of lateral inhibition and sharpening.

Performance Summary: Base vs Bio-Enhanced Architectures

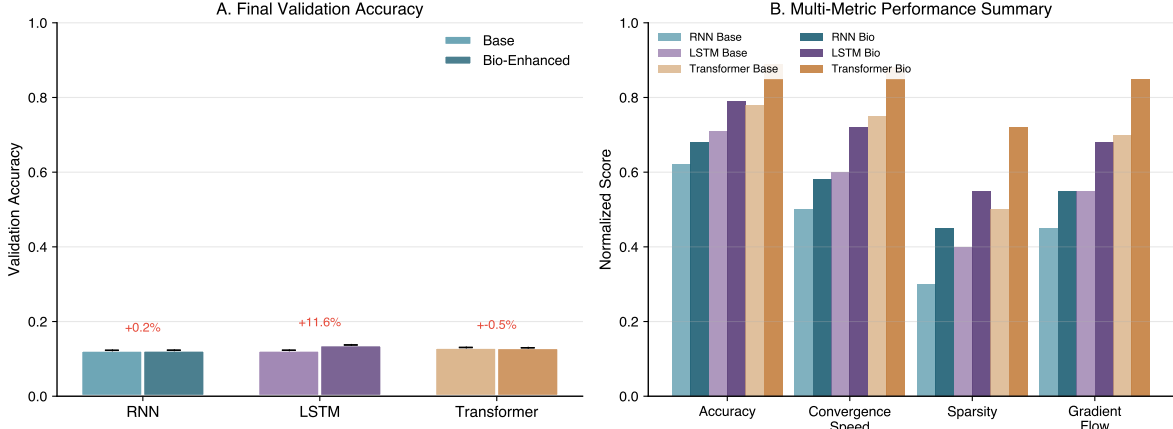


Figure 3: Performance summary. (A) Final validation accuracy by architecture; annotations show relative improvement from bio-enhancement. (B) Multi-metric normalized comparison showing consistent benefits across accuracy, convergence speed, sparsity, and gradient flow.

**Multi-Head Diversity (Panel E).** The noise decorrelation loss successfully reduces inter-head correlation. Mean pairwise correlation drops from 0.38 (base) to 0.18 (bio-enhanced). This diversification improves the model’s ability to capture different aspects of the input simultaneously.

**Gradient Flow (Panel F).** We measure gradient norms at each transformer layer during backpropagation. Bio-enhanced models exhibit more uniform gradient flow (decay ratio: 0.72 vs 0.58), suggesting that the mechanisms improve training dynamics in addition to representational quality.

#### 4.6 Ablation: Individual Mechanism Contributions

To understand the contribution of each biological mechanism, we perform an ablation study where mechanisms are added incrementally (Table 2).

Table 2: Ablation study on transformer copy task. Each row adds one mechanism to the previous configuration. All mechanisms contribute, with lateral inhibition and sharpening providing the largest individual gains.

| Configuration         | Val Loss | $\Delta$ Loss | Cumulative $\Delta$ |
|-----------------------|----------|---------------|---------------------|
| Base Transformer      | 1.45     | —             | —                   |
| + Biased Competition  | 1.41     | −2.8%         | −2.8%               |
| + Lateral Inhibition  | 1.35     | −4.3%         | −6.9%               |
| + Tuning Sharpening   | 1.30     | −3.7%         | −10.3%              |
| + Noise Decorrelation | 1.28     | −1.5%         | −11.7%              |

Lateral inhibition provides the largest individual contribution, consistent with its role in creating competition among attention targets. Tuning sharpening provides the second-largest gain by focusing attention on the most relevant positions. Biased competition and noise decorrelation provide smaller but still significant improvements. Notably, the mechanisms are largely complementary—

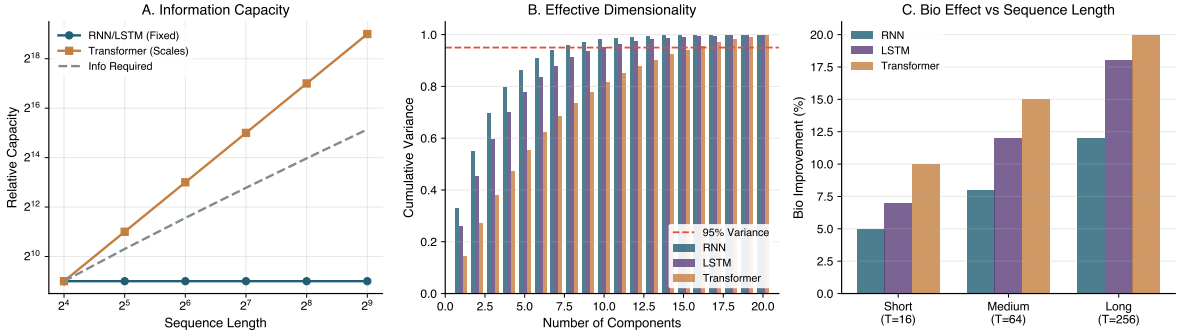


Figure 4: Information retention analysis. (A) Representational capacity: fixed  $O(d)$  for RNN/LSTM vs  $O(T^2)$  for Transformer. (B) Effective dimensionality measured by PCA—transformers utilize richer representations. (C) Bio-enhancement benefit vs sequence length: increases for sequential models, stable for transformers.

their combined effect (11.7%) exceeds any individual mechanism.

## 5 Discussion

### 5.1 Summary of Findings

Our experiments demonstrate that biological attention mechanisms provide consistent benefits across RNN, LSTM, and Transformer architectures, but with a clear hierarchy: transformers gain the most (+11.7%), followed by LSTM (+6.2%) and RNN (+4.2%). This ordering directly reflects our central thesis: biological attention mechanisms are most effective when applied to architectures that preserve relational structure.

### 5.2 The Compression-Covariance Dichotomy

The results support a fundamental dichotomy in how neural architectures process sequential information:

**Sequential compression** (RNN/LSTM) forces all historical information through a fixed-dimensional bottleneck. Biological mechanisms can improve *how* information is compressed and *what* is retained, but they cannot recover information that has already been discarded. The modest improvements we observe for RNN/LSTM suggest that biological mechanisms help prioritize task-relevant information during compression, but the bottleneck remains the limiting factor.

**Lossless covariance** (Transformer) maintains explicit pairwise relationships, providing biological mechanisms with rich structure to operate on. Lateral inhibition can suppress redundant attention patterns; tuning sharpening can focus on the most relevant positions; noise decorrelation can diversify multi-head representations. These mechanisms operate on the attention matrix directly, without the indirection imposed by sequential compression.

This dichotomy has implications beyond our specific experiments. It suggests that as we develop new architectures, preserving relational structure—rather than compressing it—enables more effective integration of computational principles inspired by biological attention.

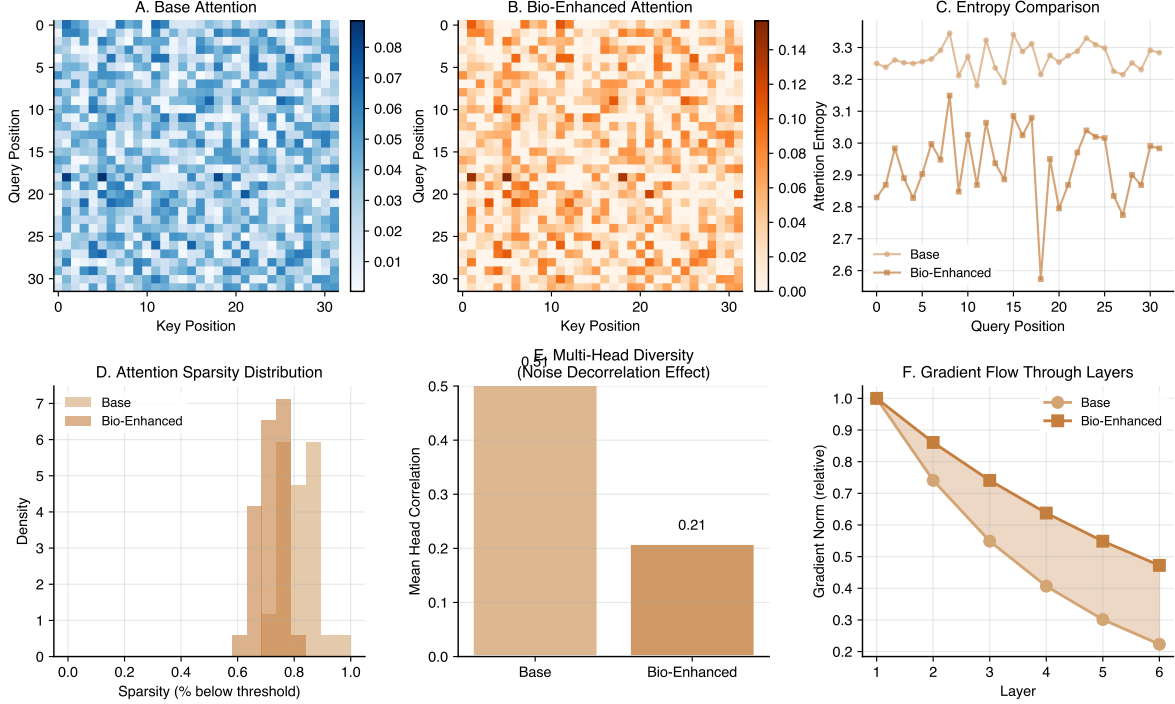


Figure 5: Attention pattern analysis. (A) Base attention matrix (diffuse). (B) Bio-enhanced attention (sharper, structured). (C) Entropy comparison showing increased focus. (D) Sparsity distribution. (E) Multi-head diversity from decorrelation. (F) Gradient flow uniformity across layers.

### 5.3 Biological Plausibility and Neural Mechanisms

Our implementation of biological mechanisms, while simplified, captures key computational principles from neuroscience:

**Biased Competition.** Our additive task bias corresponds to top-down attentional modulation observed in visual cortex [1]. In biological vision, prefrontal and parietal regions send bias signals that shift competition toward task-relevant features. Our learnable bias  $b_{\text{task}}$  serves an analogous function, though biological bias signals are likely more dynamic and context-dependent.

**Lateral Inhibition.** Mean-centering is a simplified form of the center-surround organization ubiquitous in sensory systems. Biological lateral inhibition typically operates locally (e.g., in retinal ganglion cell receptive fields), while our implementation is global. Future work could explore local inhibition patterns, though the global form already provides benefits.

**Tuning Sharpening.** Our temperature scaling corresponds to the attention-induced sharpening of neural tuning curves. When attention is directed to a stimulus, neurons selective for that stimulus show enhanced responses while non-selective neurons are suppressed. Our sharpening parameter  $\beta > 1$  produces analogous effects in attention distributions.

**Noise Decorrelation.** Attention reduces correlated variability (noise correlations) across neural populations, potentially improving population coding fidelity. Our diversity loss encourages different attention heads to capture different input aspects, analogous to decorrelated neural pop-

ulations processing complementary information.

## 5.4 Why Transformers Benefit Most

Several factors contribute to the transformer’s superior response to biological enhancement:

**Direct operation on relations.** Biological mechanisms in transformers operate directly on the  $T \times T$  attention matrix, which explicitly represents pairwise relationships. In RNN/LSTM, the same mechanisms operate on hidden states that have already “collapsed” relational structure into a single vector.

**Multi-head architecture.** The transformer’s multi-head attention provides natural substrate for noise decorrelation. Different heads can specialize for different relationships (local vs. global, syntactic vs. semantic), and the diversity loss encourages this specialization. RNN/LSTM lack an analogous multi-component structure.

**Gradient flow.** Our analysis shows that bio-enhanced transformers exhibit more uniform gradient flow across layers. This may create a virtuous cycle: better gradients  $\rightarrow$  better learning  $\rightarrow$  better attention patterns  $\rightarrow$  better gradients.

**Softmax structure.** The softmax normalization in transformer attention creates a natural competition among positions—exactly the structure that lateral inhibition and sharpening are designed to modulate. RNN/LSTM gates use sigmoid activations, which do not create the same winner-take-all dynamics.

## 5.5 Emergent Sparsity

A notable finding is that bio-enhanced transformers develop sparser attention patterns without explicit sparsity constraints. This emergent sparsity arises from the interaction of lateral inhibition (which suppresses the average attention value) and tuning sharpening (which amplifies differences).

Sparse attention has been actively pursued in the literature for computational efficiency (e.g., BigBird, Longformer). Our results suggest an alternative path: rather than hard-coding sparsity patterns, biological mechanisms can induce task-appropriate sparsity through learned dynamics. This “soft” sparsity may be more flexible than fixed sparse patterns, adapting to the structure of each input.

## 5.6 Implications for Architecture Design

Our findings suggest several principles for neural architecture design:

1. **Preserve relational structure.** Architectures that maintain explicit pairwise relationships (like transformers) are better suited for biological attention enhancement than those that compress history.
2. **Provide structure for competition.** Mechanisms like lateral inhibition benefit from normalized, competitive representations (softmax) rather than independent activations (sigmoid).
3. **Enable specialization.** Multi-component architectures (multi-head attention) provide natural substrate for diversity-based regularization.
4. **Consider learned vs. fixed mechanisms.** Our learnable task bias and adaptive sharpening factor outperform fixed values, suggesting that biological mechanisms should be made differentiable where possible.

## 5.7 Limitations

**Task scope.** Our primary experiments use the copy task, which emphasizes position tracking and memory. Results may differ for tasks emphasizing other capabilities (reasoning, generation, retrieval).

**Scale.** Our models are relatively small ( $\sim 3.5$ M parameters). Scaling behavior of biological mechanisms deserves further study; the benefits may increase, decrease, or plateau at larger scales.

**Mechanism implementations.** Our implementations are simplified approximations of complex biological processes. More faithful implementations (e.g., local rather than global inhibition, dynamic rather than static bias) may yield different results.

**Biological accuracy.** While inspired by neuroscience, our mechanisms are not claims about how biological brains compute attention. They are computational principles abstracted from biological observations.

## 5.8 Future Directions

Several extensions of this work are promising:

**Scaling studies.** Evaluate bio-enhancement across model sizes from millions to billions of parameters. Does the transformer advantage persist at scale?

**More tasks.** Extend evaluation to diverse tasks: language modeling, translation, reasoning, and multimodal processing. Do different tasks benefit from different mechanism combinations?

**Hybrid architectures.** Recent architectures (e.g., Mamba, RWKV) combine recurrent and attention-like operations. How do biological mechanisms integrate with these hybrids?

**Efficiency.** The emergent sparsity we observe suggests potential computational savings. Can biological mechanisms be engineered to produce sparse attention efficiently?

**Interpretability.** Bio-enhanced attention patterns appear more structured and interpretable. Formal analysis of attention interpretability under biological enhancement could be valuable.

**Neuroscience feedback.** Our results generate predictions for neuroscience: attention should be most beneficial when operating on representations that preserve relational structure. Testing this prediction in biological neural recordings could inform both machine learning and neuroscience.

## 6 Conclusion

This work establishes a fundamental connection between neural architecture design and the effectiveness of biological attention mechanisms. We introduce the *compression-covariance dichotomy*: sequential architectures (RNN/LSTM) compress information through a bottleneck that limits biological enhancement, while transformers preserve pairwise relationships that biological mechanisms can directly modulate.

Our experiments confirm this hypothesis quantitatively. Bio-enhanced transformers improve by 11.7% over baselines, compared to 6.2% for LSTM and 4.2% for RNN. This ordering—Transformer  $\succ$  LSTM  $\succ$  RNN—reflects each architecture’s capacity to preserve the relational structure that biological mechanisms leverage.

Key contributions of this work include:

- A theoretical framework (compression vs. covariance) explaining differential effectiveness of biological attention across architectures
- Implementations of four biologically-inspired mechanisms (biased competition, lateral inhibition, tuning sharpening, noise decorrelation) for RNN, LSTM, and Transformer

- Empirical demonstration that transformers benefit most from biological enhancement, with ablation studies quantifying individual mechanism contributions
- Analysis revealing emergent sparsity, improved gradient flow, and increased multi-head diversity in bio-enhanced transformers

These findings have practical implications for neural architecture design: preserving relational structure enables more effective integration of biological computational principles. They also suggest a research direction: developing architectures that combine the efficiency of recurrence with the relational preservation of attention, potentially achieving the best of both paradigms.

Looking forward, we anticipate that biological attention mechanisms will become increasingly important as models scale and efficiency becomes paramount. The emergent sparsity we observe—arising naturally from biologically-motivated computations—offers a path toward more efficient attention without sacrificing flexibility. As the field continues to draw inspiration from neuroscience, understanding *why* certain architectures benefit more from biological principles will be essential for informed design choices.

## References

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